

# Machine Learning for Natural Language Understanding

Prof. Achim Rettinger – Master NLP





# ML Basics

## Chapter 1



# Sources: Lecture and Exercise

- Learning Paradigms, AI vs ML vs DL
  - <https://youtu.be/chfwJiXvBMA>
- Supervised Learning (Dataset + Algorithm = Predictive Model; Linear Regression Model)
  - <https://youtu.be/UQFIss3pu-k>
- Formalization Dataset (Attributes vs. Features, discrete vs continuous, classification vs regression)
  - <https://youtu.be/PK6IEeAWiRg>
- Algorithm: Model Class, Objective, Optimizer
  - <https://youtu.be/PED1OYvQUX4>
- Text Classification (Features Generation from text)
  - [https://youtu.be/y4\\_g\\_Ni3bGw](https://youtu.be/y4_g_Ni3bGw)

# Flipped Classroom Questions

- What are differences between reinforcement learning and supervised learning?
- What is a “loss function”? Name two examples.
- What is “x” und “y” in a “supervised learning dataset”?
- What is a “bag of words”-Representation?